

Carrageenan induces cell cycle arrest in human intestinal epithelial cells in vitro.



Multiple studies in animal models have shown that the commonly used food additive carrageenan (CGN) induces inflammation and intestinal neoplasia. We performed the first studies to determine the effects of CGN exposure on human intestinal epithelial cells (IEC) in tissue culture and tested the effect of very low concentrations (1-10 mg/L) of undegraded, high-molecular weight CGN. These concentrations of CGN are less than the anticipated exposure of the human colon to CGN from the average Western diet. In the human colonic epithelial cell line NCM460 and in primary human colonic epithelial cells that were exposed to CGN for 1-8 d, we found increased cell death, reduced cell proliferation, and cell cycle arrest compared with unexposed control cells. After 6-8 d of CGN exposure, the percentage of cells reentering G0-G1 significantly decreased and the percentages of cells in S and G2-M phases significantly increased. Increases in activated p53, p21, and p15 followed CGN exposure, consistent with CGN-induced cell cycle arrest. Additional data, including DNA ladder, poly ADP ribose polymerase Western blot, nuclear DNA staining, and activities of caspases 3 and 7, indicated no evidence of increased apoptosis following CGN exposure and were consistent with CGN-induced necrotic cell death. These data document for the first time, to our knowledge, marked adverse effects of low concentrations of CGN on survival of normal human IEC and suggest that CGN exposure may have a role in development of human intestinal pathology.